Profile C — Highly Regulated / Air-Gapped Modernizer

Focus: Data sovereignty · On-prem / hybrid · Deterministic latency · Fortress-grade security

Fortress · v1.0

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COMPUTE / PLATFORM LAYER

runs disconnected · FIPS-validated

Red Hat OpenShift	RHEL + Podman	OpenShift GitOps (ArgoCD)	IBM MQ / Solace PubSub+	Temporal (self-hosted)	Next.js (self-hosted)
Enterprise K8s (on-prem/hybrid)	Hardened OS · containers	Declarative deployments	Enterprise messaging	Durable workflows	Containerized SSR
air-gap install · FIPS 140-2	DISA STIG compliant	disconnected registry mirror	guaranteed delivery · on-prem	no SaaS dependency	air-gapped build pipeline
deterministic P99 latency	OS-level patch audit trail	100% reproducible builds	no data leaves perimeter	workflow loss = 0	no external CDN required

INTEGRATION / DATA LAYER

sovereign · deterministic · auditable

IBM MQ / Solace PubSub+	Oracle DB / Postgres EE	ClickHouse (self-hosted)	Redis Enterprise	MinIO / IBM Storage	MuleSoft / IBM ACE
Event streaming	Enterprise RDBMS	On-prem OLAP	HA cache	S3-compatible on-prem	Integration layer
guaranteed once-delivery	TDE · Data Guard · RAC	sub-second · no cloud egress	active-active replication	erasure coded · on-prem	B2B protocols
audit-grade message log	RPO 0 possible	disconnected-friendly	no data leaves DC	full data sovereignty	SWIFT · HL7 · EDI

SECURITY LAYER

FIPS 140-2 · HSM-backed · air-gapped

IBM Security Verify	Vault Enterprise + HSM	CyberArk PAM	Splunk ES / QRadar	Tenable / Qualys	IBM Guardium
Identity governance	Secrets + PKI	Privileged access	SIEM	Vuln scanning	Database activity monitoring
SAML · OIDC · SCIM	Thales Luna HSM	session recording	correlation rules	STIG · CIS benchmarks	DLP · encryption
on-prem deployment	FIPS 140-2 L3 root of trust	regulator-auditable	on-prem log retention	continuous compliance	PCI-DSS evidence

OBSERVABILITY LAYER

on-prem · deterministic · disconnected

IBM Instana	Dynatrace / Splunk	OpenTelemetry Collector	Nagios / Zabbix	CMMI quantitative process mgmt
APM · auto-instrumented	Enterprise observability	Vendor-neutral SDK	Infrastructure monitoring	Statistical process control
on-prem · 1s granularity	on-prem cluster	hybrid telemetry	legacy + modern on-prem	KPI metrics framework · CAR
FIPS · air-gapped	retention policy controlled	escape hatch to A or B	SNMP · WMI · IPMI	continuous improvement loop

CMMI LEVEL 5 TRADE-OFF (Profile C)

Selected for: **compliance + determinism** . Every component runs inside a controlled perimeter; no data leaves the DC; FIPS 140-2 validated at every layer. Sacrifices: velocity (change approval gates); cost (roughly 3x Profile A per unit of workload).

Quantitative basis: P99 latency variance < 5% (deterministic hardware); 100% data-residency evidence for regulators; zero-trust HSM root of trust validated by 3 independent audits.

Reversibility condition: if regulatory perimeter relaxes OR cloud-hybrid becomes acceptable†' migrate the analytics plane only to Profile A (never the identity or audit planes, which must stay sovereign).